Spatio-Temporal Momentum — Backtest Report

Model: STM (TFT)
Period: 2026-06-01 to 2026-06-08 (6 sessions)
Universe (46): AFG, AFL, AJG, BAC, C, CBSH, CFR, CHCO, CPF, CVBF, FITB, GL, IBCP, INDB, JEF, JPM, KEY, MCO, MCY, MKL, NTRS, CNO, PNC, SBCF, SCHW, SEIC, SIGI, WAL, SRCE, STT, TFC, TMP, TRC, TROW, TRST, UBSI, UFCS, USB, VALU, VLY, WABC, WFC, WRB, ZION, HBAN, PNFP

Layout follows Tan, Roberts & Zohren (2023), arXiv:2302.10175.

Notes:
- DMN (reference model of the paper) is not implemented in this project and is not reported.
- TSMOM / CSMOM use the longest configured return horizon (21 days); the paper uses 252 days.
- Rescaled results target 15% annualised volatility with an ex-ante 20-day EWM estimate.
- Rolling correlations use a 20-session window (paper: 1 year).
- SHAP values: Expected Gradients (shap.GradientExplainer formulation); the paper uses Deep SHAP.

Table 1: Performance Metrics for Strategies – Raw Signal Outputs

	E[Return]	Vol.	Downside Deviation	MDD	Sharpe	Sortino	Calmar	Hit Rate	Ave. P / Ave. L
Long Only	1.141	0.151	0.068	0.010	7.546	16.801	110.924	0.667	1.605
TSMOM	-0.028	0.042	0.027	0.006	-0.653	-1.020	-4.853	0.333	1.794
MACD	0.111	0.031	0.017	0.003	3.578	6.513	43.499	0.500	1.818
CSMOM	-0.034	0.009	0.007	0.001	-3.825	-4.775	-27.386	0.333	1.089
STM (TFT)	0.938	0.068	0.006	0.001	13.738	149.088	966.202	0.833	4.801

Table 2: Performance Metrics for Strategies – Rescaled to Target Volatility

	E[Return]	Vol.	Downside Deviation	MDD	Sharpe	Sortino	Calmar	Hit Rate	Ave. P / Ave. L
Long Only	–	–	–	–	–	–	–	–	–
TSMOM	–	–	–	–	–	–	–	–	–
MACD	–	–	–	–	–	–	–	–	–
CSMOM	–	–	–	–	–	–	–	–	–
STM (TFT)	–	–	–	–	–	–	–	–	–

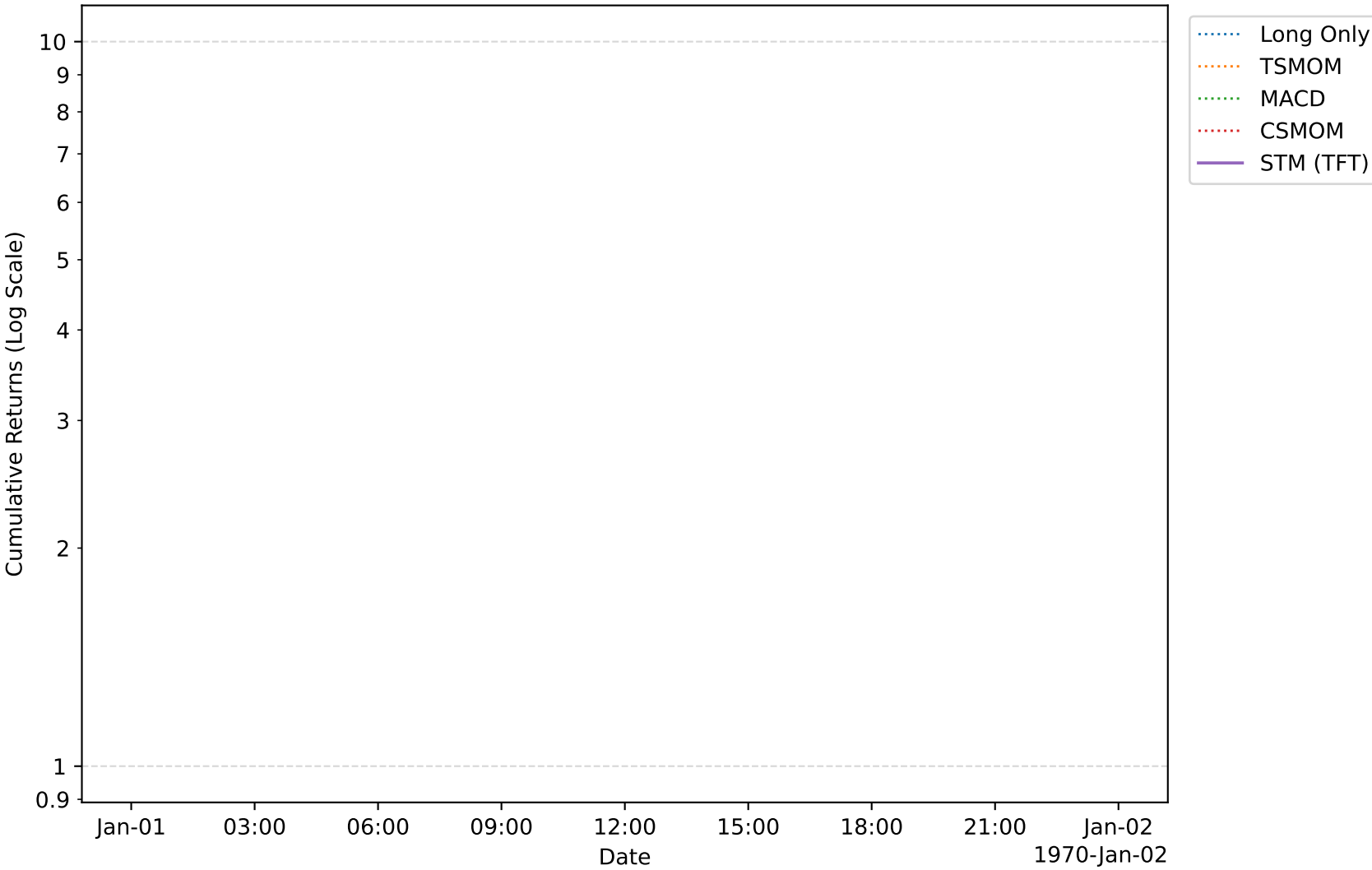

Figure 1: Cumulative Returns - Rescaled to Target Volatility
Cumulative Returns (Log Scale)
10
9
8
7
6
5
4
3
2
1
0.9
Jan-01
03:00
06:00
09:00
12:00
15:00
18:00
21:00
Jan-02
Date
1970-Jan-02
Long Only
TSMOM
MACD
CSMOM
STM (TFT)

Figure 3: (Left) Correlation of Strategies (Right) Rolling Correlations

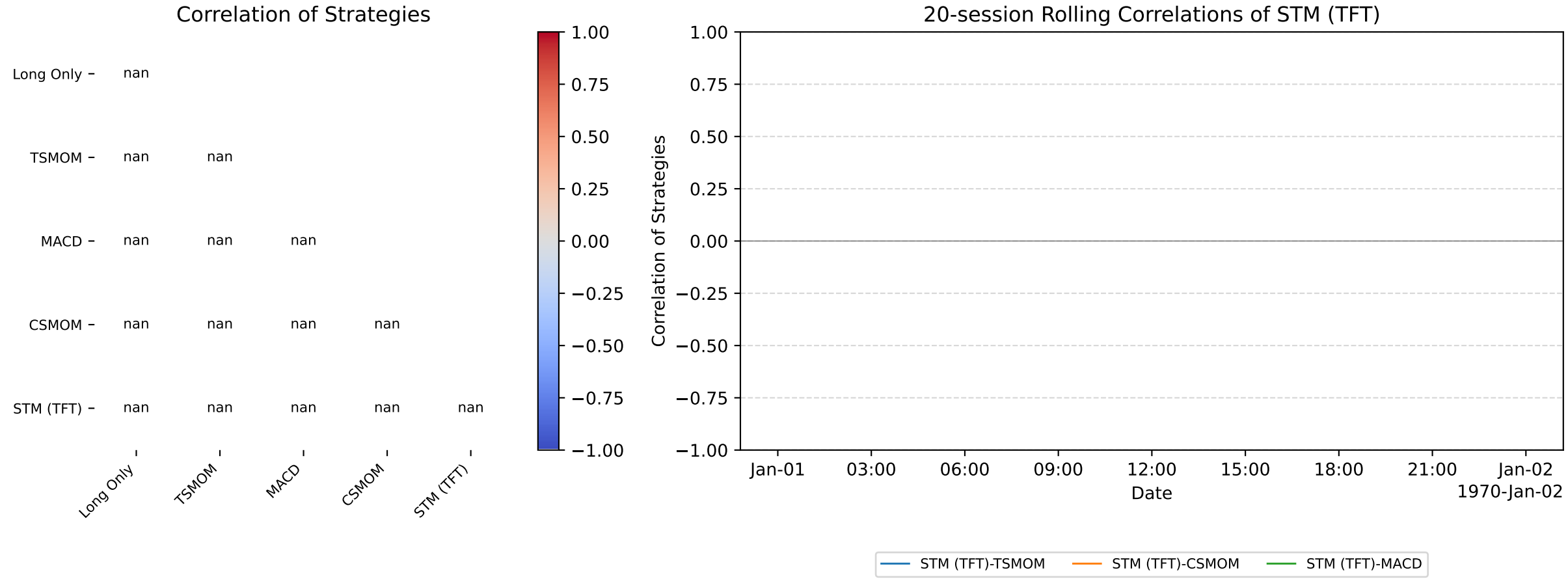
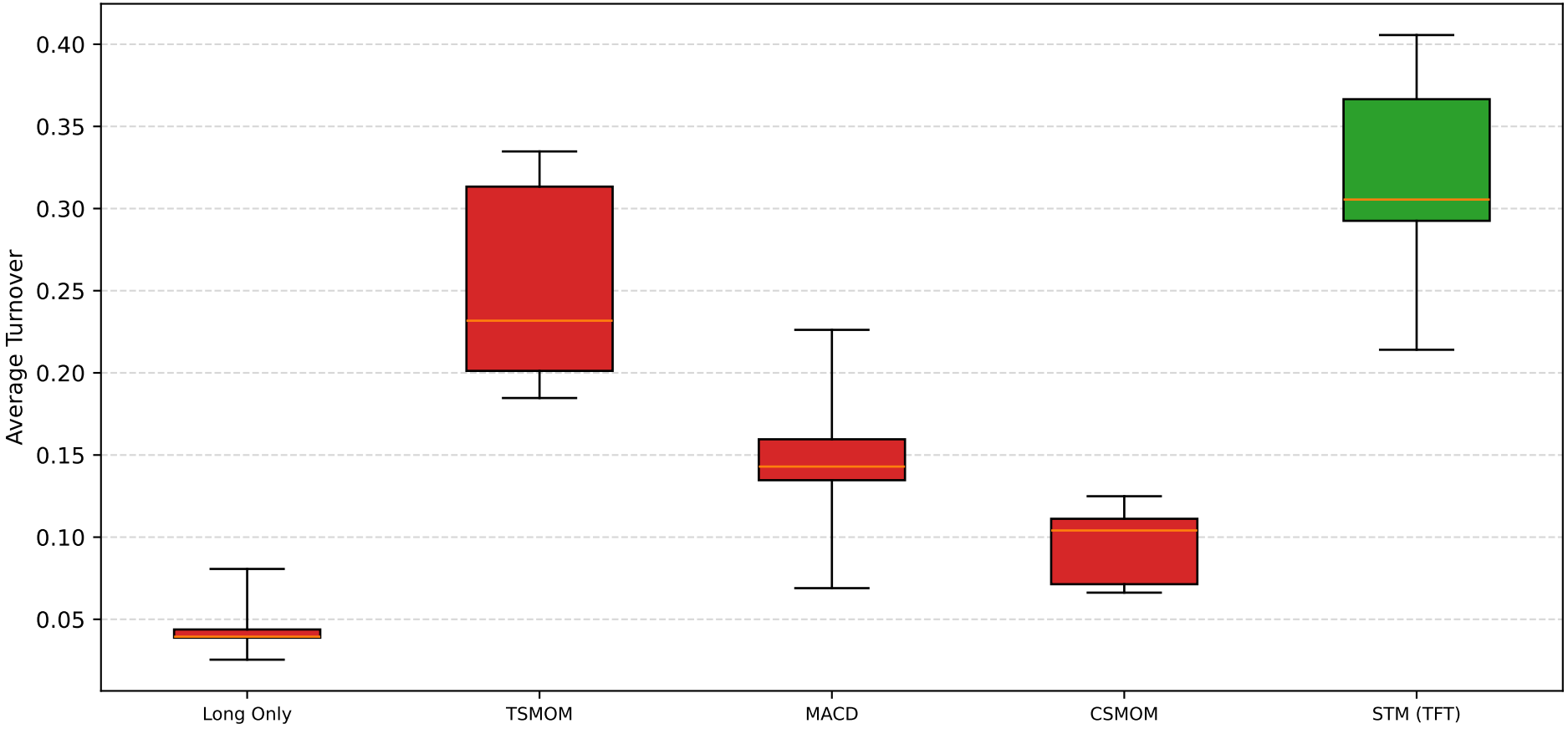


Table 6: Impact of Transactions Costs on Sharpe Ratio – Rescaled to Target Volatility (columns: transaction cost in basis points)

	0	0.5	1	2	3	4	5	10
Long Only	–	–	–	–	–	–	–	–
TSMOM	–	–	–	–	–	–	–	–
MACD	–	–	–	–	–	–	–	–
CSMOM	–	–	–	–	–	–	–	–
STM (TFT)	–	–	–	–	–	–	–	–

Figure 4: Average Turnover of Strategies



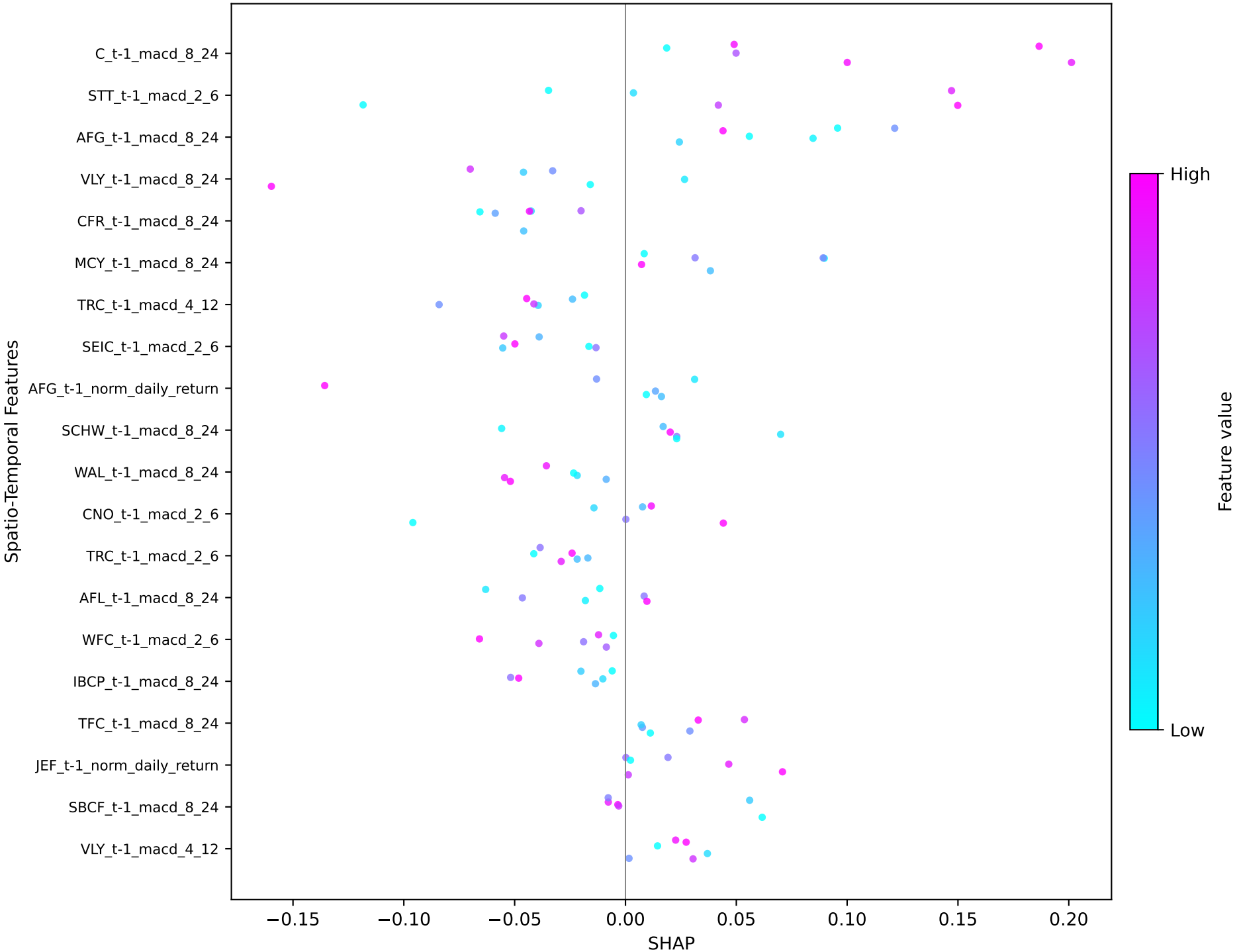

Figure 5: Impact on Predicted Signal Output for a Single Asset – BAC
C_t-1_macd_8_24
STT_t-1_macd_2_6
AFG_t-1_macd_8_24
VLY_t-1_macd_8_24
CFR_t-1_macd_8_24
MCY_t-1_macd_8_24
TRC_t-1_macd_4_12
SEIC_t-1_macd_2_6
AFG_t-1_norm_daily_return
SCHW_t-1_macd_8_24
WAL_t-1_macd_8_24
CNO_t-1_macd_2_6
TRC_t-1_macd_2_6
AFL_t-1_macd_8_24
WFC_t-1_macd_2_6
IBCP_t-1_macd_8_24
TFC_t-1_macd_8_24
JEF_t-1_norm_daily_return
SBCF_t-1_macd_8_24
VLY_t-1_macd_4_12
Spatio-Temporal Features
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
SHAP
High
Low
Feature value

Figure 6: Cumulative Mean Absolute Impact on Predicted Signal Output for All Assets

